2.1 The Challenge of Domain Shift and Unsupervised Adaptation

The widespread deployment of supervised deep learning models in precision agriculture is fundamentally limited by the "domain shift" phenomenon, wherein algorithms trained on a specific source dataset suffer precipitous declines in accuracy when applied to novel, unseen target environments (Li et al., 2025). In real-world agricultural settings, field conditions are characterized by extreme temporal and spatial variability; factors such as shifting illumination, disparate soil backgrounds, varying crop growth stages, and different sensor platforms (e.g., transitioning from high-resolution ground-based cameras to lower-resolution unmanned aerial vehicles) drastically alter the visual distribution of the acquired data (Gao et al., 2024). For example, significant detection accuracy drops of up to 14.5% have been observed simply when models are deployed across different growing seasons, underscoring how temporal distribution shifts degrade model reliability (Deng et al., 2024). Because standard Convolutional Neural Networks (CNNs) tend to memorize the localized, specific environmental statistics of their training data, they fail to generalize to these cross-crop, cross-season, or cross-continental variations, necessitating advanced adaptation strategies that do not rely on the exhaustive and costly re-annotation of the target domains (Magistri et al., 2023).

To overcome this generalization barrier, researchers have developed various Unsupervised Domain Adaptation (UDA) strategies, frequently employing adversarial learning and generative networks to enforce distribution alignment between disparate fields. One prominent approach operates at the input level by utilizing Generative Adversarial Networks (GANs) to perform contrastive style transfer, which effectively translates the visual characteristics and textures of the source images to match the target domain while strictly preserving the underlying semantic layout (Magistri et al., 2023). Alternatively, feature-level adaptation employs domain discriminators within the network architecture to compel the feature extractor to learn environment-invariant representations. For instance, multi-level attention-based adversarial discriminators have been engineered to bridge severe visual gaps, successfully adapting models trained on highly diverse, internet-sourced plant images to the specific, localized data captured by specialized agricultural robots (Tzouras et al., 2024). Other feature-level approaches combine adversarial optimization with unsupervised entropy minimization to confidently align deep features across diverse seeding beds and vegetation densities, thereby directly reducing the model's predictive uncertainty on the unlabeled target data (Ilyas et al., 2023).

Beyond generative and feature-level alignment, output-level adaptation via self-training and pseudo-labeling has emerged as a highly effective, albeit challenging, technique to bridge the domain gap. This method relies on using the base model to predict labels for the target domain, which are subsequently utilized as pseudo-ground-truth data for iterative co-training (Huang & Bais, 2024). However, severe agricultural domain shifts often induce "pseudolabel drift," a phenomenon where early misclassifications—such as confusing morphologically similar crops and weeds—are continuously reinforced throughout the training loop (Nadeem et al., 2026). To mitigate this degradation, advanced UDA frameworks integrate rigorous filtering and hybrid alignment strategies. For instance, greedy pseudo-labeling algorithms actively monitor class covariance to optimize the proportion of selected pseudo-labels, actively preventing the model from overfitting to false predictions (Huang & Bais, 2024). Similarly, recent cross-crop adaptation frameworks employ masked adversarial alignment to restrict adversarial learning strictly to low-confidence regions, such as complex crop-weed boundaries, ensuring that well-aligned, high-confidence pseudo-labels remain uncorrupted during the adaptation process (Nadeem et al., 2026).

While these UDA methodologies—ranging from GAN-driven style translation to covariance-guided pseudo-labeling—mark a substantial paradigm shift toward label-efficient agricultural vision, critical limitations in cross-domain transfer persist. Traditional global feature alignment techniques, such as standard Maximum Mean Discrepancy (MMD), can inadvertently blur inter-class boundaries and induce negative transfer by improperly forcing the alignment of distinct crop categories that are merely spectrally similar (Li et al., 2025). To counteract this, modern architectures are increasingly adopting class-aware alignment strategies that perform fine-grained adaptation within individual crop categories, successfully bridging extreme environmental gaps such as cross-continental deployments (Li et al., 2025). Furthermore, despite utilizing robust implicit instance-invariant networks (I3Net) to mitigate structural differences (Deng et al., 2024), extreme domain shifts often cause "cold-start" failures where the baseline model cannot generate a sufficient volume of reliable pseudo-labels to successfully initiate the self-training loop (Li et al., 2025). Consequently, achieving universal generalization across disparate agricultural domains requires carefully synthesizing structural style transfer, feature-level adversarial alignment, and rigorous, class-aware pseudo-label filtering.

2.2 Zero-Shot Learning, Foundation Models, and Label Efficiency

The development of robust computer vision systems for precision agriculture has historically been bottlenecked by the prohibitive cost and labor associated with pixel-level manual annotation. Standard supervised deep learning models require massive, finely annotated datasets to generalize across diverse field conditions, a requirement that is economically and logistically intractable for real-world agricultural scaling (Ghose et al., 2025). To circumvent this dependency, the field is undergoing a significant paradigm shift toward label-efficient learning and visual foundation models. Rather than training architectures from scratch on dense annotations, researchers are increasingly adopting self-supervised representations, such as DenseCL and MoCo v2, which learn spatial features directly from unlabeled plant imagery to reduce downstream annotation needs (Ogidi et al., 2026). Simultaneously, massive pre-trained foundation models, including Vision-Language Models (VLMs) and the Segment Anything Model (SAM), are being deployed to achieve open-vocabulary, zero-shot segmentation, enabling agricultural robots to delineate plant boundaries without requiring task-specific fine-tuning or explicit labels (Chong et al., 2024).

Despite the generalization capabilities of foundation models, deploying them in a strictly zero-shot manner often yields semantic confusion in chaotic agricultural environments, prompting a shift toward hybrid, prompt-guided architectures. Because pure SAM lacks the inherent botanical semantics to differentiate crops from weeds, it frequently over-segments backgrounds or mistakenly categorizes crops as weeds when unguided (Soniya et al., 2024). To resolve this, researchers are integrating lightweight object detectors with foundation models to create unified detection-segmentation frameworks. For instance, the YOLO-SAM AgriScan framework utilizes a YOLOv11 detector, fine-tuned via few-shot learning, to generate precise bounding box prompts that guide SAM2 in performing zero-shot pixel-level segmentation of ripe strawberries (Ghose et al., 2025). Similarly, Grounded YOLO-SAM architectures pair YOLOv10 with advanced SAM variants (SAM2 and SAM2.1) to extract highly accurate, temporally consistent masks of cotton bolls across various developmental stages. This cooperative methodology effectively merges the rapid localization capabilities of YOLO with the zero-shot segmentation power of SAM, drastically reducing annotation requirements while maintaining high spatial fidelity for tasks like weed density estimation (Soniya et al., 2024).

In parallel to foundation models, label efficiency is being advanced through the use of synthetic data generation and semi-supervised pseudo-labeling. Generative AI pipelines have successfully combined SAM's zero-shot masking capabilities with fine-tuned, text-to-image Stable Diffusion models to synthesize highly realistic artificial training images. This approach demonstrates that substituting up to 90% of real-world training data with synthetic images can actually improve downstream detection metrics on edge devices by introducing beneficial variability (Modak & Stein, 2024). Furthermore, semi-supervised object detection (SSOD) frameworks leverage a teacher-student paradigm to exploit vast pools of unlabeled field data. Frameworks such as WeedTeacher integrate dynamic confidence calibration and Mixup/Mosaic augmentations within a YOLOv8 architecture to generate and refine pseudo-labels for unlabeled vegetable field images, achieving superior accuracy over fully supervised baselines using only 10% of the labeled data (Deng et al., 2025). Similar semi-supervised approaches applied to FCOS and Faster R-CNN architectures have proven capable of reaching up to 96% of their fully supervised detection accuracy while drastically cutting manual labeling costs (Li et al., 2024).

However, despite their immense potential, these foundation models and label-efficient paradigms exhibit critical limitations and distinct failure cases under severe field conditions. In zero-shot anomaly segmentation, where weeds are classified as anomalous plants compared to a crop baseline, the models fail catastrophically during severe weed infestations because the anomalous weeds overwhelm the field and become the visual majority, breaking the algorithm's foundational assumption (Chong et al., 2024). Hybrid YOLO-SAM pipelines are highly susceptible to cascading errors; because SAM relies entirely on the detector's prompts, any failure by YOLO to localize an occluded or visually ambiguous target results in a complete segmentation miss by the foundation model (Ghose et al., 2025). Furthermore, generative diffusion models carry the risk of producing unverified misclassifications or introducing data leakage if the synthetic patterns inadvertently overlap with test distributions (Modak & Stein, 2024). Finally, while semi-supervised pseudo-labeling excels in-domain, it struggles severely under cross-domain shifts; when models encounter completely new fields or imaging platforms, the teacher network generates inaccurate high-confidence pseudo-labels, resulting in minimal to zero performance improvement over standard supervised baselines (Deng et al., 2025).

2.3 Edge Deployment and Identification of the Research Gap

Although Vision Transformers (ViTs) and visual foundation models exhibit superior global spatial dependency modeling, their massive parameter counts and quadratic computational complexity pose critical barriers to real-time deployment on resource-constrained agricultural edge devices, such as unmanned aerial vehicles (UAVs) and field robots (Luca, 2025). The substantial memory and power requirements of these architectures severely limit drone battery flight durations and onboard processing speeds, precipitating a conflict between algorithmic complexity and deployment feasibility (Luca, 2025). To reconcile the necessity for high-resolution contextual modeling with stringent latency and power budgets, researchers are increasingly engineering lightweight hybrid architectures that amalgamate the global reasoning of ViTs with the localized, computationally efficient feature extraction of Convolutional Neural Networks (CNNs) (Yenna et al., 2026). For example, compact CNN-ViT hybrids incorporating convolutional attention blocks have been demonstrated to achieve high classification accuracies with minimal memory footprints, ensuring real-time processing capabilities on embedded hardware (Yenna et al., 2026; Galymzhankyzy & Martinson, 2025). Additional model compression techniques, such as mixed-precision quantization, operator fusion, and layer-adaptive magnitude-based pruning, are concurrently employed to accelerate inference speeds without sacrificing the detection recall of fine-grained agricultural targets (Mo et al., 2025).

Despite these architectural optimizations, systematic landscape reviews of agricultural artificial intelligence reveal several persistent, open challenges that hinder the commercial scalability of autonomous weeding systems. Foremost among these is a pervasive lack of rigorous, real-world field validation; fewer than 30% of developed systems are evaluated in uncontrolled agricultural conditions, leaving the majority of models highly fragile to unpredictable outdoor lighting, weather fluctuations, and topological variations (Hamrani et al., 2025). Furthermore, comprehensive reviews consistently highlight poor cross-scenario generalization, wherein models suffer precipitous performance declines when confronting novel soil types, diverse imaging platforms, or unseen weed communities, largely due to a reliance on narrow datasets and flawed validation practices that permit spatial or temporal data leakage (Hamrani et al., 2025; Abedin & Mehrub, 2026). Compounding these generalization failures is a documented weakness in current multimodal processing; while multi-sensor data holds immense theoretical promise, existing fusion systems struggle with severe spatio-temporal synchronization errors, resolution disparities, and a lack of adaptive weighting mechanisms to effectively fuse heterogeneous data streams during real-time inference (Hamrani et al., 2025).

The convergence of these hardware constraints and generalization limitations explicitly defines the current research gap in precision agricultural vision. While Foundation Models and zero-shot learning paradigms offer a transformative solution to the prohibitive bottleneck of pixel-level manual annotation, they still exhibit catastrophic failures under severe field conditions, particularly when morphologically similar weeds and crops densely overlap in standard RGB imagery (Abedin & Mehrub, 2026; Luca, 2025). Relying solely on unimodal RGB optical data restricts perception to surface-level morphological features, which are easily obscured by canopy overlap, dynamic illumination, and hard shadows, inevitably leading to critical semantic confusion between target crops and anomalous weeds (Luca, 2025). Therefore, the necessary next step for achieving robust, universal weed detection is the strategic fusion of Multimodal sensor data—incorporating thermal, multispectral, and depth (RGB-D) information—with the cognitive and zero-shot segmentation power of Foundation Models (Hamrani et al., 2025; Galymzhankyzy & Martinson, 2025). Integrating these heterogeneous sensory inputs into a unified, computationally efficient vision-language or transformer-based framework will enable autonomous weeding systems to transcend the visual ambiguities of purely optical data, leveraging deep structural and spectral insights to reliably identify unseen weed species across diverse, real-world agricultural ecosystems (Abedin & Mehrub, 2026).